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Question 2

Let show that :

2a) All eigenvalues of an invertible matrix are non-zero

Let A be an invertible matrix and λ be an eigenvalue of A with eigenvector $v \neq 0$.
By definition, the eigenvalues of A are the roots of the characteristic polynomial:

$$\det(A - \lambda I) = 0$$

If $\lambda = 0$ is an eigenvalue, then:

$$\det(A) = 0$$

But this contradicts the fact that A is invertible (invertible matrices have non-zero determinant).
Therefore, all eigenvalues must be non-zero.

2b) If λ is an eigenvalue of invertible A , then $1/\lambda$ is an eigenvalue of A^{-1}

Let λ be an eigenvalue of A with eigenvector $v \neq 0$:

$$Av = \lambda v$$

Since A is invertible, we can multiply both sides by A^{-1} from the left:

$$A^{-1}(Av) = A^{-1}(\lambda v)$$

$$v = \lambda A^{-1}v$$

Dividing both sides by λ (which is non-zero by part 2a):

$$\frac{1}{\lambda}v = A^{-1}v$$

or equivalently:

$$A^{-1}v = \frac{1}{\lambda}v$$

According to the definition, $\frac{1}{\lambda}$ is the eigenvalue of A^{-1}

2c) If λ is an eigenvalue of A , then λ^2 is an eigenvalue of A^2

Let λ be an eigenvalue of A with eigenvector $v \neq 0$:

$$Av = \lambda v$$

So,

$$A^2v = A(Av) = A(\lambda v) = \lambda(Av) = \lambda(\lambda v) = \lambda^2v$$

Therefore:

$$A^2v = \lambda^2v$$

This shows that v is an eigenvector of A^2 with eigenvalue λ^2 .

Question 3

Let $A_1, A_2 \in \mathbb{R}^{n \times n}$ with spectra $\sigma(A_1) = \{\lambda_1, \dots, \lambda_n\}$ and $\sigma(A_2) = \{\delta_1, \dots, \delta_n\}$.

In the general case, where A_1 and A_2 do **not** necessarily commute (i.e., $A_1A_2 \neq A_2A_1$), the spectrum of the sum $A_1 + A_2$ and the product A_1A_2 **cannot be determined** solely from the individual spectra $\sigma(A_1)$ and $\sigma(A_2)$.

The eigenvalues of the sum and product depend on the relationship between the eigenvectors of the two matrices. For arbitrary matrices, there is no fixed algebraic relationship between the individual eigenvalues and the eigenvalues of their sum or product.

If A_1 and A_2 are Commuting Matrices

$$A_1A_2 = A_2A_1$$

then A_1 and A_2 are **simultaneously triangularizable**. This implies that there exists an ordering of the eigenvalues such that:

$$\sigma(A_1 + A_2) = \{\lambda_i + \delta_i \mid i = 1, \dots, n\}$$

$$\sigma(A_1A_2) = \{\lambda_i\delta_i \mid i = 1, \dots, n\}$$

Question 4

Given symmetric $A \in \mathbb{R}^{n \times n}$ with spectrum $\sigma(A) = \{\lambda_1, \lambda_2, \dots, \lambda_n\}$.

4a) Value of γ for $B = A + \gamma I$ to be positive semi-definite

A symmetric matrix is positive semi-definite if and only if all its eigenvalues are non-negative.

Since A is symmetric, it has real eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$.

The eigenvalues of $B = A + \gamma I$ are:

$$\sigma(B) = \{\lambda_1 + \gamma, \lambda_2 + \gamma, \dots, \lambda_n + \gamma\}$$

We need:

$$\lambda_i + \gamma \geq 0 \quad \text{for all } i = 1, 2, \dots, n$$

This means:

$$\gamma \geq -\lambda_i \quad \text{for all } i$$

Therefore:

$$\boxed{\gamma \geq -\min_{i=1, \dots, n} \lambda_i = -\lambda_{\min}}$$

where $\lambda_{\min}$ is the smallest eigenvalue of A .

4b) Lower bound for γ given specific spectrum

Given: $\sigma(A) = \left\{ (-1)^k \frac{n^2 - k + 1}{n^2 + k} : k = 1, \dots, n \right\}$

For $B = A + \gamma I$ to be positive definite, we need:

$$\lambda_i + \gamma > 0 \quad \text{for all } i$$

Therefore:

$$\gamma > -\lambda_{\min}$$

We need to find $\lambda_{\min}$ from the given spectrum.

For $k = 1, 2, \dots, n$, we have:

$$\lambda_k = (-1)^k \frac{n^2 - k + 1}{n^2 + k}$$

- When k is odd: $\lambda_k = -\frac{n^2 - k + 1}{n^2 + k} < 0$
- When k is even: $\lambda_k = \frac{n^2 - k + 1}{n^2 + k} > 0$

The minimum eigenvalue occurs at the largest odd k .

- If n is odd: $k = n$ gives $\lambda_n = -\frac{n^2 - n + 1}{n^2 + n} = -\frac{n^2 - n + 1}{n(n+1)}$
- If n is even: $k = n - 1$ gives $\lambda_{n-1} = -\frac{n^2 - (n-1) + 1}{n^2 + (n-1)} = -\frac{n^2 - n + 2}{n^2 + n - 1}$

General lower bound:

$$\boxed{\gamma > \frac{n^2 - n + 1}{n(n+1)}}$$

Question 5

Given: $f(x) = x_1^2 e^{(1-x_1^2)} - 20.25(x_1 - x_2)^2$

(i) Manual Computation

Gradient Computation:

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix}$$

For $\frac{\partial f}{\partial x_1}$:

$$\frac{\partial f}{\partial x_1} = \frac{\partial}{\partial x_1} \left[x_1^2 e^{(1-x_1^2)} \right] - 20.25 \frac{\partial}{\partial x_1} [(x_1 - x_2)^2]$$

$$\begin{aligned} \frac{\partial}{\partial x_1} \left[x_1^2 e^{(1-x_1^2)} \right] &= 2x_1 e^{(1-x_1^2)} + x_1^2 e^{(1-x_1^2)} \cdot (-2x_1) \\ &= 2x_1 e^{(1-x_1^2)} - 2x_1^3 e^{(1-x_1^2)} \\ &= 2x_1 e^{(1-x_1^2)} (1 - x_1^2). \end{aligned}$$

$$\begin{aligned} 20.25 \frac{\partial}{\partial x_1} [(x_1 - x_2)^2] &= 20.25 \cdot 2(x_1 - x_2) \\ &= 40.5(x_1 - x_2) \end{aligned}$$

$$\boxed{\frac{\partial f}{\partial x_1} = 2x_1 e^{(1-x_1^2)} (1 - x_1^2) - 40.5(x_1 - x_2)}$$

For $\frac{\partial f}{\partial x_2}$:

$$\frac{\partial f}{\partial x_2} = -20.25 \cdot 2(x_1 - x_2) \cdot (-1)$$

$$\boxed{\frac{\partial f}{\partial x_2} = 40.5(x_1 - x_2)}$$

At point $x = (0, -1)$:

$$\left. \frac{\partial f}{\partial x_1} \right|_{(0, -1)} = 2(0)e^1(1 - 0) - 40.5(0 - (-1)) = 0 - 40.5 = -40.5$$

$$\left. \frac{\partial f}{\partial x_2} \right|_{(0, -1)} = 40.5(0 - (-1)) = 40.5$$

$$\boxed{\nabla f(0, -1) = \begin{bmatrix} -40.5 \\ 40.5 \end{bmatrix}}$$

Hessian Computation:

$$H(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix}$$

- $\frac{\partial f}{\partial x_1} = 2x_1 e^{(1-x_1^2)}(1-x_1^2) - 40.5(x_1 - x_2)$

So,

$$\begin{aligned} \frac{\partial^2 f}{\partial x_1^2} &= 2e^{(1-x_1^2)}(1-x_1^2) + 2x_1(-2x_1)e^{(1-x_1^2)}(1-x_1^2) + 2x_1e^{(1-x_1^2)}(-2x_1) - 40.5 \\ &= 2e^{(1-x_1^2)}(1-x_1^2)(1-2x_1^2) - 4x_1^2e^{(1-x_1^2)} - 40.5 \end{aligned}$$

At $(0, -1)$:

$$\left. \frac{\partial^2 f}{\partial x_1^2} \right|_{(0,-1)} = 2e^1(1)(1) - 0 - 40.5 = 2e - 40.5 \approx -35.06$$

- $\frac{\partial^2 f}{\partial x_1 \partial x_2} = \frac{\partial}{\partial x_2}[-40.5(x_1 - x_2)] = 40.5$
- $\frac{\partial f}{\partial x_2} = 40.5(x_1 - x_2)$

So,

$$\frac{\partial^2 f}{\partial x_2^2} = \frac{\partial}{\partial x_2}[40.5(x_1 - x_2)] = -40.5$$

$$H(0, -1) = \begin{bmatrix} 2e - 40.5 & 40.5 \\ 40.5 & -40.5 \end{bmatrix} \approx \begin{bmatrix} -35.06 & 40.5 \\ 40.5 & -40.5 \end{bmatrix}$$

Question 8

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is **convex** if its Hessian matrix $H(x)$ is positive semi-definite for all x in its domain, $H(x) \succeq 0$.

8a) $f(x) = x_1^2 e^{(1-x_1^2)} - 20.25(x_1 - x_2)^2$

From Question 5:

$$H(x) = \begin{bmatrix} 2e^{(1-x_1^2)}(1-x_1^2)(1-2x_1^2) - 4x_1^2e^{(1-x_1^2)} - 40.5 & 40.5 \\ 40.5 & -40.5 \end{bmatrix}$$

$$H_{22} = -40.5 < 0 \text{ for all } x.$$

Thus, $H(x)$ has a negative eigenvalue everywhere, so f is **NOT CONVEX**

8b) $f(x) = (x_1^2 + x_2 - 11)^2 + (x_1 + x_2^2 - 7)^2$

Hessian computation:

$$\begin{aligned}\frac{\partial f}{\partial x_1} &= 2(x_1^2 + x_2 - 11)(2x_1) + 2(x_1 + x_2^2 - 7) \\ &= 4x_1(x_1^2 + x_2 - 11) + 2(x_1 + x_2^2 - 7), \\ \frac{\partial f}{\partial x_2} &= 2(x_1^2 + x_2 - 11) + 2(x_1 + x_2^2 - 7)(2x_2) \\ &= 2(x_1^2 + x_2 - 11) + 4x_2(x_1 + x_2^2 - 7).\end{aligned}$$

$$\begin{aligned}\frac{\partial^2 f}{\partial x_1^2} &= 12x_1^2 + 4x_2 - 42, \\ \frac{\partial^2 f}{\partial x_2^2} &= 12x_2^2 + 4x_1 - 26, \\ \frac{\partial^2 f}{\partial x_1 \partial x_2} &= 4(x_1 + x_2). \\ H(x) &= \begin{pmatrix} 12x_1^2 + 4x_2 - 42 & 4(x_1 + x_2) \\ 4(x_1 + x_2) & 12x_2^2 + 4x_1 - 26 \end{pmatrix}.\end{aligned}$$

Consider the point $(x_1, x_2) = (0, 0)$:

$$H(0, 0) = \begin{pmatrix} -42 & 0 \\ 0 & -26 \end{pmatrix}.$$

Both eigenvalues are negative, hence the Hessian is not positive semidefinite at $(0, 0)$.

Since the Hessian of f fails to be positive semi-definite at some point, **f is NOT CONVEX**.

8c) $f(x) = x_1^4 + x_2^4 + 2x_1^2x_2^2 + 3x_1^2 + 4x_2^2$

Gradient:

$$\nabla f = \begin{bmatrix} 4x_1^3 + 4x_1x_2^2 + 6x_1 \\ 4x_2^3 + 4x_1^2x_2 + 8x_2 \end{bmatrix}$$

Hessian:

$$H(x) = \begin{bmatrix} 12x_1^2 + 4x_2^2 + 6 & 8x_1x_2 \\ 8x_1x_2 & 12x_2^2 + 4x_1^2 + 8 \end{bmatrix}$$

- $\det H = (12x_1^2 + 4x_2^2 + 6)(12x_2^2 + 4x_1^2 + 8) - (8x_1x_2)^2$

$$\det(\mathbf{H}) = 144x_1^2x_2^2 + 48x_1^4 + 96x_1^2 + 48x_2^4 + 16x_1^2x_2^2 + 32x_2^2 + 72x_2^2 + 24x_1^2 + 48 - 64x_1^2x_2^2$$

$$\det(\mathbf{H}) = 48x_1^4 + 48x_2^4 + 96x_1^2x_2^2 + 120x_1^2 + 104x_2^2 + 48$$

Since $x_1^4, x_2^4, x_1^2x_2^2, x_1^2$, and x_2^2 are always non-negative, and all coefficients are positive, $\det(\mathbf{H}) > 0$.

Thus, $H(x) \succeq 0$ everywhere: **f is CONVEX**.